ASCII MASTER FLOW — PANEL 1/12: Exogenic source-transfer-storage system

```
+-------------- EXOGENIC LANDSCAPE SYSTEM --------------+
| CENTRAL QUESTION: How is rock converted into landform? |
+------------------------------------------------------------+
                         |
                ROCK EXPOSED AT SURFACE
                         |
                 weathering in place
                         |
            regolith and loose material supplied
              /                          \
 gravity-driven transfer          agent-driven transfer
 mass wasting downslope           water, wind, ice, waves
              \                          /
                 erosion + transport
                         |
                     deposition
                         |
                 temporary storage
                         |
                    remobilisation
```

Denudation is collective lowering; weathering, erosion and mass wasting
are linked but distinct processes.
MUST REMEMBER: Weathering prepares material in situ, mass movement transfers it downslope,
  running water removes it, and groundwater follows infiltration, percolation, aquifer storage
  and discharge; Indian analysis links lithology, slope, rainfall, land use and pumping.

ASCII MASTER FLOW — PANEL 2/12: Weathering taxonomy, controls, regolith and soil formation

```
                      WEATHERING
             +------------+------------+
             |            |            |
         MECHANICAL     CHEMICAL     BIOLOGICAL
         thermal        solution     roots/burrows
         unloading      carbonation  organic acids
         frost          hydration    human exposure
         salt           hydrolysis
         wet-dry        oxidation
             \            |            /
              +------ REGOLITH -------+
                        |
 organic inputs + organisms + translocation + time
                        |
                      SOIL
               O -> A/E -> B -> C -> R
               removal    gain     parent rock

CONTROLS CONVERGE: mineralogy + joints/bedding + climate + relief
                   + drainage + organisms + time + land use

Black soil link: Deccan basalt parent, shrink-swell clay, deep dry cracks;
parent material does not alone determine every soil property.
```

ASCII MASTER FLOW — PANEL 3/12: Soil erosion sequence and watershed control logic

```
RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION
                         |
                 diffuse sheet removal
                         |
                  small rill channels
                         |
                 entrenched gullies
                         |
             ravine landscape: Chambal cue

WATERSHED RESPONSE
cover soil -> slow runoff -> shorten slope length -> contour/terrace
-> protect drainage -> treat gully head -> rehabilitate catchment

WIND BRANCH
surface creep -> saltation -> suspension

COASTAL BRANCH
wave erosion belongs to the coastal-process system; use only when asked.

A check dam alone cannot replace upstream cover, drainage and maintenance.
```

ASCII MASTER FLOW — PANEL 4/12: Mass-movement taxonomy and slope-stability mechanism

```
                     MASS MOVEMENT
       +------------------+------------------+
       |                  |                  |
     SLOW               SLIDE/FALL          FLOW/AVALANCHE
 creep, solifluction    rockfall            debris flow
                       translational slide  earth or mud flow
                       rotational slump     rock/debris avalanche
                                            snow avalanche

SLOPE BALANCE
resisting strength = cohesion + friction + structural support
                         versus
driving stress = gravity component + loading + toe loss
                         |
rain infiltration -> added weight + pore pressure -> lower effective strength
road or river cut -> steepening / undercutting -> support loss
quake / quarry / leakage / drainage change -> trigger or amplification
                         |
                 FAILURE WHEN DRIVE > RESISTANCE

Material, movement and water define the event; speed alone does not.
```

ASCII MASTER FLOW — PANEL 5/12: India landslide geography, risk production and mitigation stack

```
INDIA LANDSLIDE GEOGRAPHY
Himalaya / NE : young relief + fractures + incision + rain + seismicity
Western Ghats : escarpment + deep weathering + orographic rain + cuts
Nilgiris      : weathered slopes + roads + settlements + drainage change
                         |
PREDISPOSITION -> TRIGGER -> MOVEMENT -> EXPOSURE -> LOSS
 geology/slope    rain/quake   slide/flow   road/town    disaster
                         |
GSI LADDER
inventory -> susceptibility/hazard study -> post-event diagnosis
-> monitoring / NLFC support -> decision communication
                         |
MITIGATION STACK
avoid unsafe siting -> manage drainage -> protect toe
-> stabilise slope -> bioengineer -> control spoil/quarrying
-> maintain works -> warn/close routes -> evacuate and prepare

Inventory maps past events; susceptibility maps relative likelihood;
neither is a deterministic event prediction.
Joshimath is a multi-causal subsidence case, not a synonym for landslide.
```

ASCII MASTER FLOW — PANEL 6/12: Groundwater architecture from recharge to wells

```
rain -> infiltration -> soil moisture -> percolation -> recharge

GROUND SURFACE      recharge area                 spring at slope
-------------------------v-------------------------------->
ZONE OF AERATION       perched water on local aquitard
-------------------- capillary fringe -------------------
WATER TABLE       UNCONFINED AQUIFER        pumping well
~~~~~~~~~~~~~~~~~~~~~~~\________________ cone of depression
AQUITARD          slow transmission
---------------------------------------------------------
CONFINED AQUIFER   pressure; water rises to potentiometric level
=========================================================
AQUICLUDE          practically no transmission

porosity = fraction of voids | permeability = connected flow capacity
aquifer stores and transmits | aquitard transmits slowly
artesian water rises under pressure; it flows only if head exceeds ground

Budget: change in storage = recharge + inflow - discharge - pumping - outflow.
```

ASCII MASTER FLOW — PANEL 7/12: India groundwater depletion, quality and subsidence geography

```
+----------------+----------------------+-------------------------+
| REGION         | AQUIFER FORM         | MAJOR PRESSURE          |
+----------------+----------------------+-------------------------+
| Indo-Gangetic  | layered alluvium     | depletion, As, nitrate  |
| Peninsular     | weathered fractures  | discontinuous storage   |
| Deccan basalt  | contacts + fractures | uneven well yield       |
| Arid hard rock | limited recharge     | fluoride, salinity      |
| Coastal belt   | fresh-saline wedge   | intrusion and upconing  |
+----------------+----------------------+-------------------------+

crop-water demand + cheap energy + private wells + urban growth
+ weak aquifer regulation + limited recharge
                         |
falling head -> lower yield -> higher lift cost -> unequal access
                         |
compressible sediment: pore-pressure loss -> compaction -> subsidence

Quality paths differ: arsenic redox | fluoride rock-water interaction
nitrate sewage/fertiliser | salinity coastal or inland
Quantity and quality require separate evidence, though each limits usable water.
```

ASCII MASTER FLOW — PANEL 8/12: Recharge, demand governance and integrated Mains answer spine

```
GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS

SUPPLY / RECHARGE                    DEMAND / GOVERNANCE
roof and runoff capture              crop-water alignment
managed aquifer recharge             efficient irrigation
watershed treatment                  well registration/metering
spring-shed restoration              conjunctive use and reuse
quality-safe source water            community water budgets
maintenance and monitoring           enforceable state/local rules
              \                          /
               +---- AQUIFER PLAN ----+
                          |
CGWB assessment + NAQUIM mapping + state regulation
+ Atal Bhujal learning + local institutions + public data
                          |
ANSWER SPINE
1 define stock-flow-quality system | 2 draw aquifer or India map
3 diagnose regional mechanism       | 4 use dated evidence carefully
5 match intervention to cause       | 6 qualify limits and equity

Recharge structures are inputs, not proof of water-table recovery.
Desalination supplies water but does not restore freshwater aquifer head.
```

ASCII MASTER FLOW — PANEL 9/12: Evidence ladder and confidence control

EVIDENCE LADDER

1. = from source book = inference / standard knowledge = current affairs.
2. Agriculture.md uses "chernozem", "alluvial" and "black soil" as...
3. Evidence units available in this file

VERDICT: Source type controls the strength and limits of the historical claim.

ASCII MASTER FLOW — PANEL 10/12: Examiner traps and contested boundaries

```
CLOSE DISTINCTIONS
1. Causal method Initial condition -> energy/force or driver ->...
2. SCALE, INTERACTING CONTROL OR EVIDENCE LIMIT
3. UPSC TRAP / ANSWER-USE: A landslide can deliver sediment to a river,...
VERDICT: Qualification earns marks; false certainty is a hard failure.
CLOSE DISTINCTION: Weathering differs from erosion, porosity from permeability, water table
  from piezometric surface, aquifer from aquitard, and slump/slide/flow/creep require
  movement-specific geometry rather than the generic label landslide.
```

ASCII MASTER FLOW — PANEL 11/12: Integrated answer spine and qualified conclusion

```
ANSWER SPINE
1. Original solved Mains questions 6
2. Local official papers/keys checked: 2018, 2019, 2020, 2021, 2023,...
3. Practice contract Every solved item has demand decoding, a detailed...
VERDICT: Claim, named evidence, analysis and qualification lead to a graded...
EVIDENCE LIMIT: A trigger such as intense rain is not the complete cause: antecedent moisture,
  discontinuities, toe cutting, drainage and construction condition failure; groundwater
  trends and notified areas require dated official status.
```

ASCII MASTER FLOW — PANEL 12/12: Evidence ladder and confidence control — DEEP-REVIEW SYNTHESIS 2

EVIDENCE LADDER

1. = from source book = inference / standard knowledge = current affairs.
2. Agriculture.md uses "chernozem", "alluvial" and "black soil" as...
3. Evidence units available in this file

VERDICT: Source type controls the strength and limits of the historical claim.